JEE Mathematics: Quick Review with Grading and Practice Map - Maxima and Minima (Batch XIII)

Grade shows how often the item is tested in the JM exercise of this batch, the JA exercise and the JEE Main Question Bank taken together (45 questions; a question that appears in two of these is counted once): High = 4 or more questions, Medium = 1 to 3 questions, Low = none. Low only means the item is not tested in these questions; it can still be important for understanding. Practice map lists the questions that test the item in the sheet exercises (J-I, J-II, J-M, J-A, C-I, C-II), the DPPs and the Question Bank (QB). 'Main concept' = the question is mainly about this item; 'Also used in' = the question needs it as a secondary step.

Maxima and Minima at a point

Question 01

What is the condition for a function to have maxima at $x = a$?

Model Answer

f has a (local) maxima at $x = a$ if $f(a) > f(a - h)$ and $f(a) > f(a + h)$ for every sufficiently small $h > 0$, i.e. $f(a)$ is the largest value in a small neighbourhood of a.

Ex: $f(x) = 1 - x^2$: $f(0) = 1 > 1 - h^2 = f(\pm h)$, so $x = 0$ is a maxima.

Grade - Low

Practice map, main concept: DPP: 5(1)

Also used in: none

Question 02

What is the condition for a function to have minima at $x = a$?

Model Answer

f has a (local) minima at $x = a$ if $f(a) < f(a - h)$ and $f(a) < f(a + h)$ for every sufficiently small $h > 0$.

Ex: $f(x) = |x|$: $f(0) = 0 < h = f(\pm h)$, so $x = 0$ is a minima.

Grade - Low

Practice map, main concept: Sheet: J-II 123

Also used in: DPP: 5(1)

Question 03

What to keep in mind while checking maxima/minima if one side of $x = a$ is not in domain?

Model Answer

Maxima and minima are points of the domain. At an end point compare $f(a)$ only with the values on the side that lies in the domain: in $[a, b]$, $x = a$ is a maxima if $f(a) > f(a + h)$, a minima if $f(a) < f(a + h)$.

Ex: $f(x) = x$ on $[0, 1]$: $x = 0$ is a minima and $x = 1$ is a maxima.

Grade - Low

Practice map, main concept: none

Also used in: Sheet: J-II 98

Question 04
What is the condition on continuity of a function to define maxima/minima?

Model Answer
None: to define maxima / minima the function need not be continuous or differentiable at $x = a$; only the values $f(a)$, $f(a \pm h)$ are compared.
Ex: $f(x) = x^2$ $(x \neq 0)$, $f(0) = 1$: discontinuous at 0, yet $x = 0$ is a maxima since $1 > h^2$.

Grade - High
Practice map, main concept: none
Also used in: Sheet: J-I 50, 52, 58, 63; J-II 119; J-A 3.02, 3.04; C-I 15 | DPP: 5(5, 17, 19) | Question Bank: 174, 178

Important Points

Question 05
What are the synonyms of maxima?

Model Answer
Maxima, maximum, local maxima, local maximum, relative maxima, relative maximum.

Grade - Low
Practice map, main concept: none
Also used in: none

Question 06
What are the synonyms of minima?

Model Answer
Minima, minimum, local minima, local minimum, relative minima, relative minimum.

Grade - Low
Practice map, main concept: none
Also used in: none

Question 07
What are the synonyms of greatest value?

Model Answer
Greatest value = global maxima / maximum = absolute maxima / maximum. On $[a, b]$ it is the largest of f at the critical points and at a, b; the value at a maxima need not be the greatest value.
Ex: $x^3 - 3x$ on $[-2, 3]$: $f(-2) = -2$, $f(-1) = 2$, $f(1) = -2$, $f(3) = 18$: greatest value 18 (local maximum value only 2).

Grade - High
Practice map, main concept: Sheet: J-I 49, 55, 57; J-II 43, 56, 58, 63, 80-82, 115, 122; J-M 11-12, 22, 30, 33, 43; J-A 3.11, 3.16; C-I 14, 22; C-II 38 | DPP: 5(11); 6(5) | Question Bank: 176, 179, 184, 187-188, 191-192
Also used in: Sheet: J-II 57

Question 08
What are the synonyms of least value?

Model Answer
Least value = global minima / minimum = absolute minima / minimum; on $[a, b]$ the smallest of f at the critical points and the end points. A local maximum value can be less than a local minimum value.
Ex: $x^3 - 3x$ on $[-2, 3]$: least value -2 (at $x = -2$ and $x = 1$).

Grade - High
Practice map, main concept: Sheet: J-I 51; J-II 44-45 | DPP: 5(8); 6(1, 4, 8-9)
Also used in: Sheet: J-I 49, 57, 62; J-II 43, 56-58, 115, 122; J-M 11-12, 33, 43; J-A 3.11, 3.16; C-I 14, 32; C-II 29 | Question Bank: 176, 179, 187, 191-192

Question 09
Define extremum points and what are its synonyms?

Model Answer
A point that is either a maxima or a minima is an extremum point; synonyms: extrema, extremum, extremal points, turning points. For a continuous non-constant function maxima and minima occur alternately.

Grade - Low
Practice map, main concept: Sheet: J-I 53
Also used in: none

First Derivative Approach

Question 10
What is the condition on $f'(x)$ in neighbourhood of point of local maxima?

Model Answer
$f'(x)$ changes sign from positive to negative as x crosses a ($f' > 0$ on the left, $f' < 0$ on the right) $\Rightarrow$ local maxima.

Grade - Low
Practice map, main concept: Sheet: C-I 15
Also used in: Sheet: J-I 72

Question 11
What is the condition on $f'(x)$ in neighbourhood of point of local minima?

Model Answer
$f'(x)$ changes sign from negative to positive as x crosses $a \Rightarrow$ local minima.

Grade - Low
Practice map, main concept: none
Also used in: none

Question 12

What if $f'(x)$ does not change its sign in neighbourhood of $x = a$?

Model Answer

Then $x = a$ is neither a maxima nor a minima (the function keeps increasing or keeps decreasing through a).

Ex: $f(x) = x^3$: $f'(0) = 0$ but $f' \geq 0$ on both sides, so $x = 0$ is not an extremum.

Grade - Medium

Practice map, main concept: DPP: 5(5-6)

Also used in: Sheet: J-M 31; J-A 3.15 | DPP: 5(12) | Question Bank: 185

Question 13

How to determine points of local maxima/minima by first derivative approach?

Model Answer

Find the domain; find the critical points ($f' = 0$ or f' does not exist, inside the domain); make the sign scheme of f'; $+ \to -$: maxima, $- \to +$: minima, no change: neither. At points where f' does not exist compare values directly.

Ex: $f'(x) = (x-1)(x-2)^2(x-3)(x-4)$: signs $-, +, +, -, +$: minima at 1, 4; maxima at 3; $x = 2$ neither.

Grade - High

Practice map, main concept: Sheet: J-I 50, 52, 54, 58-60, 62-63; J-II 59, 61, 64, 74, 91-92, 96, 98, 100, 119, 121; J-M 7, 9, 21, 31, 34, 45-47, 50; J-A 3.01-3.02, 3.04, 3.06, 3.12-3.15; C-I 18, 30, 32; C-II 29, 37, 49 | DPP: 5(2-4, 7, 10, 12-20); 6(2, 12) | Question Bank: 174-175, 177-178, 181-183, 185, 190, 195-196

Also used in: Sheet: J-I 56-57, 73; J-II 77-78, 117; J-A 3.08, 3.11; C-I 8, 28 | DPP: 6(4-5) | Question Bank: 176

Second Derivative Approach

Question 14

What can you say about $x = a$ if $f'(a) = 0$ and $f''(a)$ is positive?

Model Answer

$f'(a) = 0$, $f''(a) > 0 \Rightarrow x = a$ is a point of local minima.

Ex: $f = 4x^3 - 12x$: $f' = 12(x-1)(x+1)$, $f'' = 24x$; $f''(1) = 24 > 0$: minima at $x = 1$.

Grade - Medium

Practice map, main concept: Sheet: J-II 95; J-A 3.10 | DPP: 6(3)

Also used in: Sheet: J-II 96 | DPP: 5(2)

Question 15

What can you say about $x = a$ if $f'(a) = 0$ and $f''(a)$ is negative?

Model Answer

$f'(a) = 0$, $f''(a) < 0 \Rightarrow x = a$ is a point of local maxima.

Ex: same f: $f''(-1) = -24 < 0$: maxima at $x = -1$.

Grade - Low

Practice map, main concept: none

Also used in: Sheet: J-II 59

Question 16
What can you say about $x = a$ if $f'(a) = 0$ and $f''(a) = 0$?

Model Answer
The second derivative test fails; use the first derivative test (sign change of f') or the nth derivative test.

Ex: x^4 (minima at 0) and x^3 (no extremum at 0) both have $f'(0) = f''(0) = 0$.

Grade - Low
Practice map, main concept: none
Also used in: DPP: 5(9)

nth derivative test

Question 17
How to find points of maxima/minima with the help of nth derivative test?

Model Answer
If $f'(c) = f''(c) = \cdots = f^{(n-1)}(c) = 0$ and $f^{(n)}(c) \neq 0$: n even: $f^{(n)}(c) < 0$ maxima, $f^{(n)}(c) > 0$ minima; n odd: no extremum (f increasing about c if $f^{(n)}(c) > 0$, decreasing if < 0).

Ex: $x^5 - 5x^4 + 5x^3 - 1$ at $x = 0$: $f' = f'' = 0$, $f'''(0) = 30$ ($n = 3$ odd): neither.

Grade - Low
Practice map, main concept: DPP: 5(9)
Also used in: none

Significance on sign of second order derivative

Question 18
What is the condition on $f''(x)$ if its graph is concave up?

Model Answer
$f''(x) > 0$ on $(a, b) \Rightarrow y = f(x)$ is concave up there (the sheet calls it 'concave'; some books 'convex').

Ex: e^x, x^2 are concave up everywhere.

Grade - Medium
Practice map, main concept: Sheet: J-II 76, 118 | DPP: 6(18)
Also used in: Sheet: J-A 3.05-3.06

Question 19
What is the condition on $f''(x)$ if its graph is concave down?

Model Answer
$f''(x) < 0$ on $(a, b) \Rightarrow y = f(x)$ is concave down there (the sheet calls it 'convex').

Ex: $\ln x$ ($f'' = -\frac{1}{x^2} < 0$) is concave down on $(0, \infty)$.

Grade - Low
Practice map, main concept: DPP: 6(17)
Also used in: none

Question 20
What is the location of tangent if graph of function is concave up/down?

Model Answer
Concave up: every tangent lies below the curve; concave down: every tangent lies above the curve.
Ex: tangent to e^x at $x = 0$ is $y = 1 + x$, so $e^x \geq 1 + x$ for all x.

Grade - Low
Practice map, main concept: none
Also used in: Sheet: J-II 75

Question 21
What is the nature of $f'(x)$ if graph of $f(x)$ is concave up?

Model Answer
$f'' > 0 \Rightarrow f'(x)$ is increasing (slope of the tangent increases).

Grade - Low
Practice map, main concept: none
Also used in: Sheet: J-II 76-78, 118

Question 22
What is the nature of $f'(x)$ if graph of $f(x)$ is concave down?

Model Answer
$f'' < 0 \Rightarrow f'(x)$ is decreasing.

Grade - Low
Practice map, main concept: none
Also used in: Sheet: J-II 77-78 | DPP: 6(17)

Question 23
How to prove an inequality (or find a maximum) using concavity (chord and curve)?

Model Answer
Concave up ($f'' > 0$): every chord lies above the curve, so $f\left(\frac{x_1+x_2}{2}\right) < \frac{f(x_1)+f(x_2)}{2}$ (also for weights, e.g. $\frac{2x_1+x_2}{3}$); concave down: reverse. To compare f with g on an interval with equal end values, use that one is concave down and the other concave up.
Ex: $\sin x$ is concave down on $(0, \pi)$: $\frac{\sin A+\sin B+\sin C}{3} \leq \sin\frac{A+B+C}{3}$, so in a triangle $\sin A + \sin B + \sin C \leq \frac{3\sqrt{3}}{2}$.

Grade - Medium
Practice map, main concept: Sheet: J-II 75; J-A 3.05
Also used in: none

Point of Inflection

Question 24
Define point of inflection.

Model Answer
A point where the graph has a tangent line and the concavity changes (concave up $\leftrightarrow$ concave down).
Ex: $y = x^3$ at $x = 0$ ($f'' = 6x$ changes sign).

Grade - Low
Practice map, main concept: none
Also used in: Sheet: J-II 61, 74 | DPP: 6(10)

Question 25

What is the value of $f''(x)$ at point of inflection?

Model Answer

$f''(x) = 0$ or $f''(x)$ does not exist, and f'' must change sign there. $f'' = 0$ alone is not enough.

Ex: $y = x^{1/3}$: $f''(0)$ does not exist, sign changes: inflection. $y = x^4$: $f''(0) = 0$, no sign change: not inflection.

Grade - Low

Practice map, main concept: Sheet: J-I 72; J-II 117; C-I 28 | DPP: 6(10)

Also used in: none

Question 26

What is the relation of inflection points of $y = f(x)$ with $f'(x)$?

Model Answer

Points of inflection of $y = f(x)$ are extremum points of $y = f'(x)$ (f' changes from increasing to decreasing or vice versa).

Ex: $f = x^3 - 3x^2$: $f' = 3x^2 - 6x$ has its minimum at $x = 1$, which is the inflection point of f.

Grade - Low

Practice map, main concept: Sheet: J-II 77-78

Also used in: Sheet: J-II 60; C-I 31

Geometrical Problems

Question 27

What are the steps to solve maximizing/minimizing questions on geometrical problems?

Model Answer

Draw a figure and separate constants from variables; write the quantity to be maximised / minimised in one variable using the given relation (Pythagoras, similar triangles, fixed volume / perimeter); find its extrema by f' and also check the end values of the allowed interval.

Ex: rectangle in a semicircle of radius r: $A = x\sqrt{r^2 - \frac{x^2}{4}}$, maximum at $x = r\sqrt{2}$, $A_{\max} = r^2$.

Grade - High

Practice map, main concept: Sheet: J-I 64-68, 70-71; J-II 65-73; J-M 10, 13, 20, 32, 44, 48-49; J-A 3.07, 3.09; C-I 16, 19, 21; C-II 27-28, 30, 32-36, 39, 41 | DPP: 6(6-7, 11, 14-16, 19) | Question Bank: 180, 186, 194

Also used in: none

Analysis of Cubic Equation

Question 28

What is the process to find number of real/imaginary roots of a cubic equation?

Model Answer

$f = ax^3 + bx^2 + cx + d$, $f' = 3ax^2 + 2bx + c$. If D of $f' = 0$ is ≤ 0: f monotonic, one real root (three equal roots when $D = 0$ and f vanishes at the root of f'). If $D > 0$ with roots α, β: $f(\alpha)f(\beta) > 0$: one real, two imaginary; $= 0$: two coincident + one; < 0: three distinct real roots.

Ex: $x^3 - 3x + a = 0$: three distinct real roots iff $f(1)f(-1) = (a-2)(a+2) < 0$, i.e. $-2 < a < 2$.

Grade - Medium

Practice map, main concept: Sheet: J-I 56; J-A 3.08; C-I 8 | DPP: 6(20) | Question Bank: 193

Also used in: none

Question 29
What is the condition for positive point of maxima for a cubic polynomial?

Model Answer
For $a > 0$ the maxima is at the smaller root α of $f' = 0$ and the minima at the larger root β (swap if $a < 0$). Positive point of maxima ($a > 0$): both roots of $f' = 0$ distinct and positive: $D > 0$, $-\frac{b}{3a} > 0$, $f'(0) = c > 0$.
Ex: $x^3 - 6x^2 + 9x$: $f' = 3(x-1)(x-3)$: maxima at $x = 1 > 0$.

Grade - Low
Practice map, main concept: Sheet: J-II 79
Also used in: Sheet: J-I 54

Question 30
What is the condition for negative point of maxima for a cubic polynomial?

Model Answer
($a > 0$) The smaller root of $f' = 0$ is negative: $D > 0$ and $[f'(0) < 0$, or $-\frac{b}{3a} < 0]$.
Ex: $x^3 + 3x^2$: $f' = 3x(x+2)$: maxima at $x = -2$.

Grade - Medium
Practice map, main concept: Sheet: J-M 8 | Question Bank: 189
Also used in: none

Question 31
What is the condition for positive point of minima for a cubic polynomial?

Model Answer
($a > 0$) The larger root of $f' = 0$ is positive: $D > 0$ and $[f'(0) < 0$, or $-\frac{b}{3a} > 0]$.
Ex: $x^3 - 3x$: $f' = 3(x-1)(x+1)$: minima at $x = 1$.

Grade - Medium
Practice map, main concept: none
Also used in: Sheet: J-M 8 | Question Bank: 189

Question 32
What is the condition for negative point of minima for a cubic polynomial?

Model Answer
($a > 0$) Both roots of $f' = 0$ distinct and negative: $D > 0$, $-\frac{b}{3a} < 0$, $f'(0) > 0$.
Ex: $x^3 + 6x^2 + 9x$: $f' = 3(x+1)(x+3)$: minima at $x = -1$.

Grade - Low
Practice map, main concept: none
Also used in: none

Question 33
How to find point of inflection for a cubic polynomial?

Model Answer
$f'' = 6ax + 2b = 0 \Rightarrow x = -\frac{b}{3a}$ (always one inflection point; it is the mid-point of the abscissae of the maxima and minima and the centre of symmetry of the graph).
Ex: $x^3 - 3x^2$: $f'' = 6x - 6 = 0$, inflection at $x = 1$ (between $x = 0$ and $x = 2$).

Grade - Low
Practice map, main concept: Sheet: J-I 73-74 | DPP: 6(13)
Also used in: Sheet: J-II 60; C-I 31

Miscellaneous

Question 34

How to find a polynomial when its points of extrema (and some values or a limit) are given?

Model Answer

Write $f'(x) = k(x-\alpha)(x-\beta)\dots$ (least degree) and integrate, or write the general polynomial; a limit such as $\lim_{x\to 0} \frac{f(x)}{x^m}$ finite fixes the low coefficients; 'f' has an extremum at c' means $f''(c) = 0$. Then use the given values.

Ex: cubic with maxima 3 at $x = -1$, minima -1 at $x = 1$: $f = k\left(\frac{x^3}{3} - x\right) + C$ gives $k = 3$, $C = 1$: $f = x^3 - 3x + 1$.

Grade - **Medium**

Practice map, main concept: Sheet: J-I 61; J-II 57, 60, 62; J-A 3.03; C-I 31; C-II 31

Also used in: none